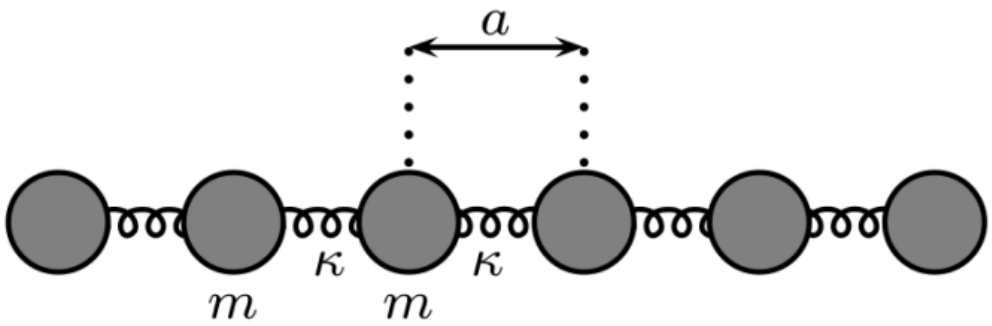
Physical Chemistry of Materials

Lecture 12

Crystal Vibrations: Phonons

Last Class

Normal Modes of 1D Crystal

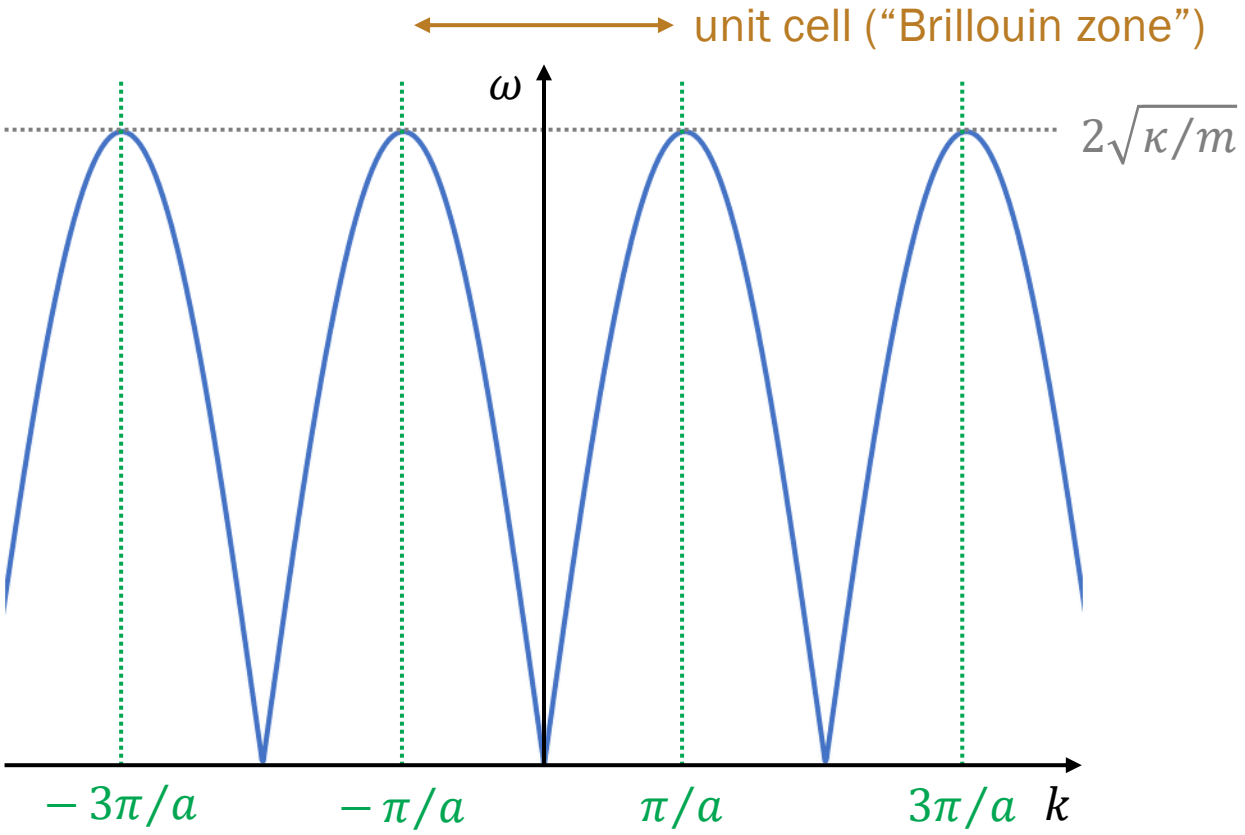


- Deviation from equilibrium position $q_n = x_n - x_{n0}$
- Total potential energy

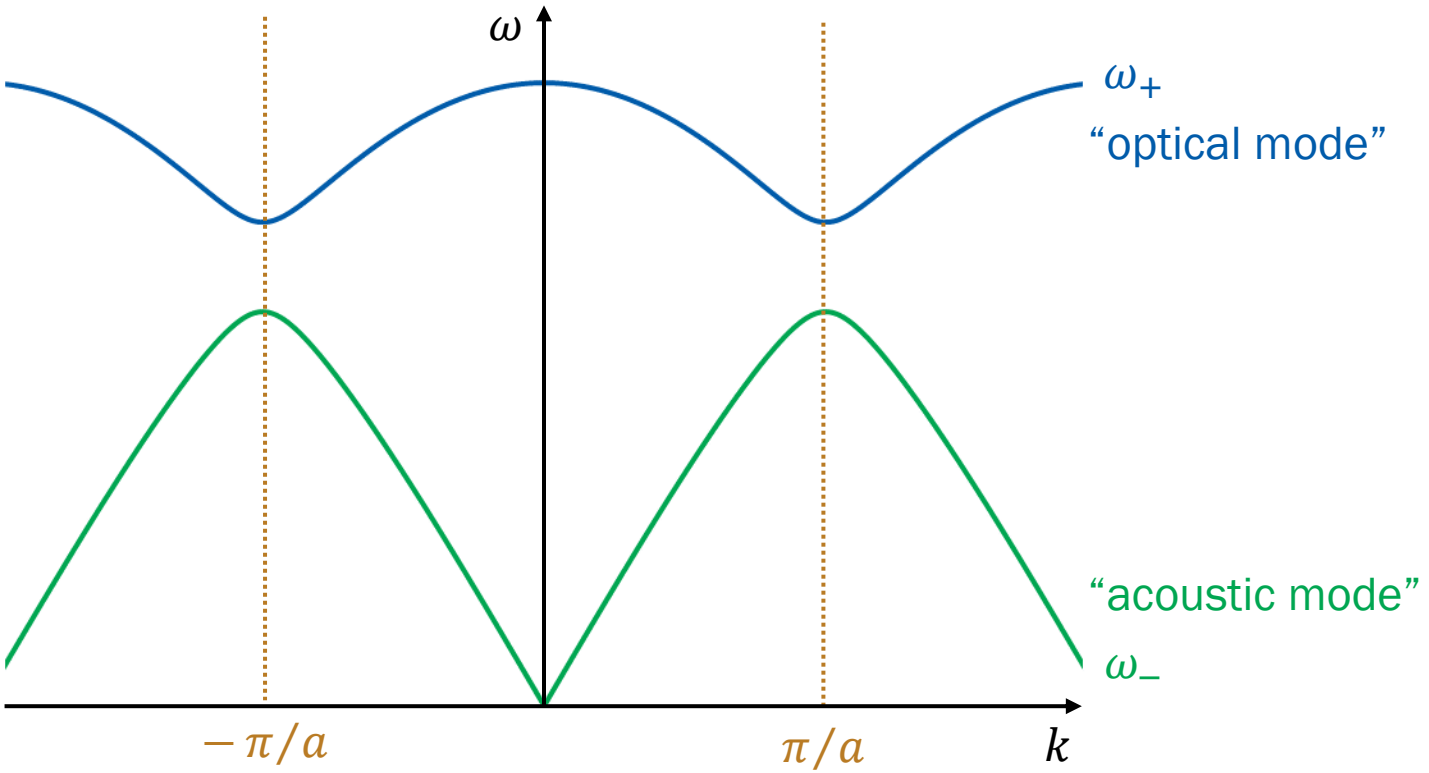
$$V_{\text{tot}} = \sum_{i=1}^{N} \frac{\kappa}{2}(x_{i+1} - x_i - a)^2 = \sum_{i=1}^{N} \frac{\kappa}{2}(q_{i+1} - q_i)^2$$

(Image: Simon, *The Oxford Solid State Basics* (2013), Fig. 9.1)

Dispersion Relation

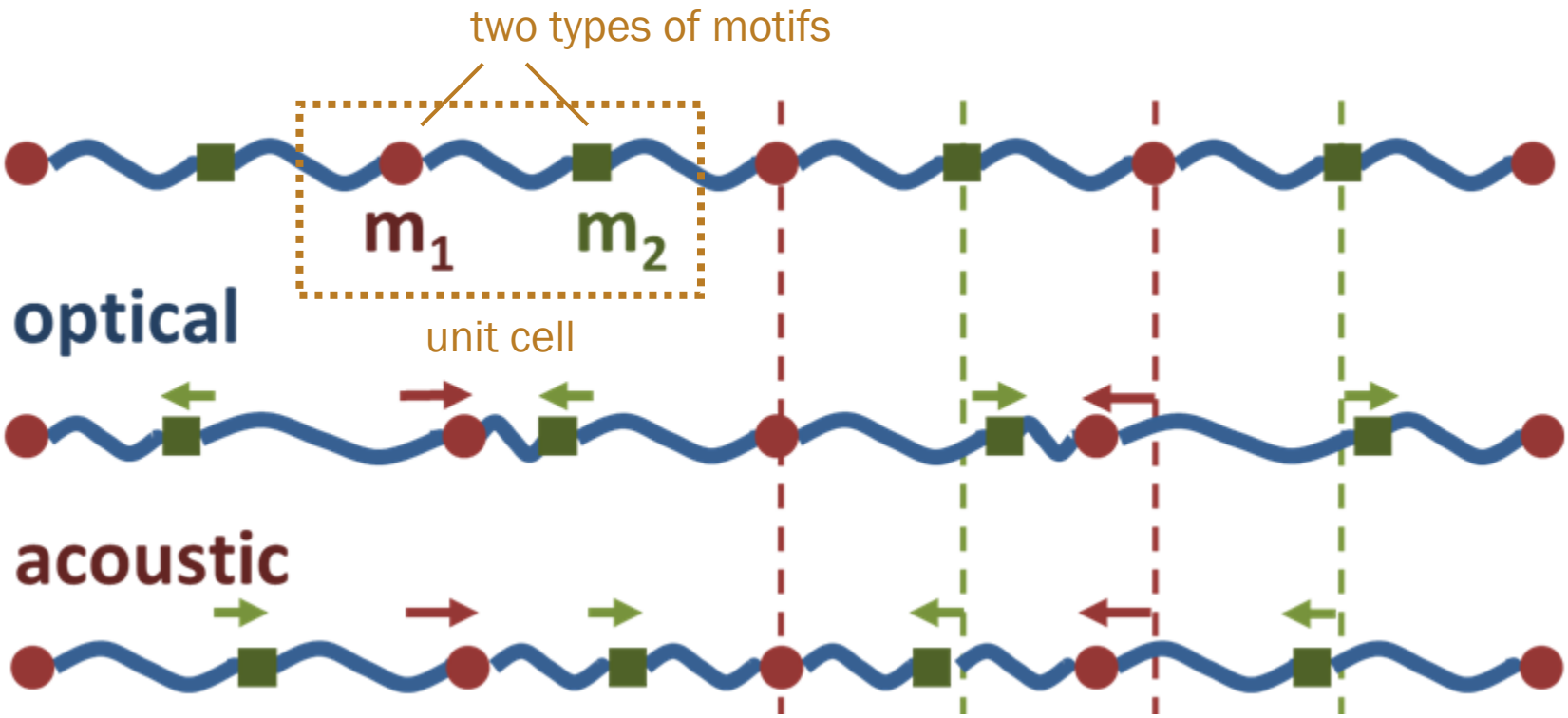


Case of Diatomic Solid



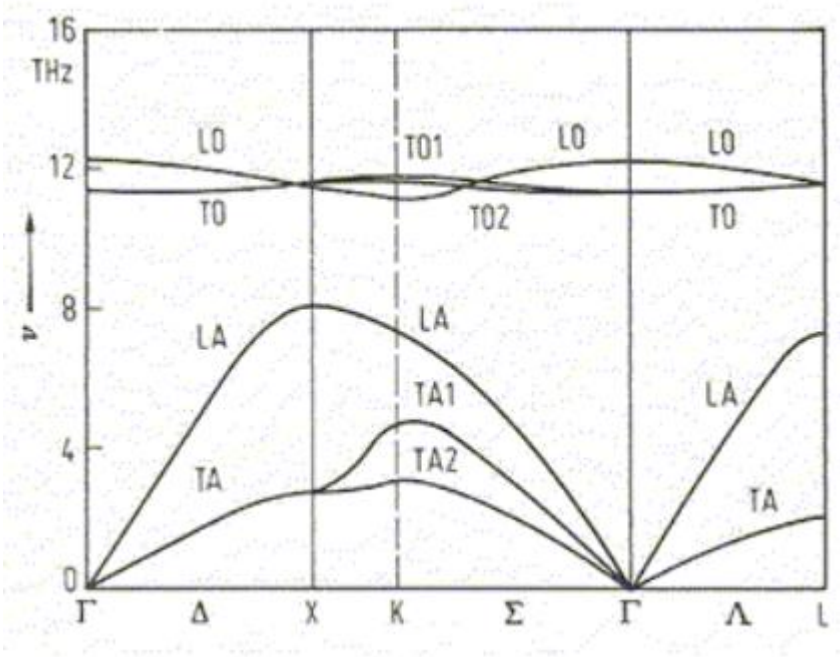
Last Class

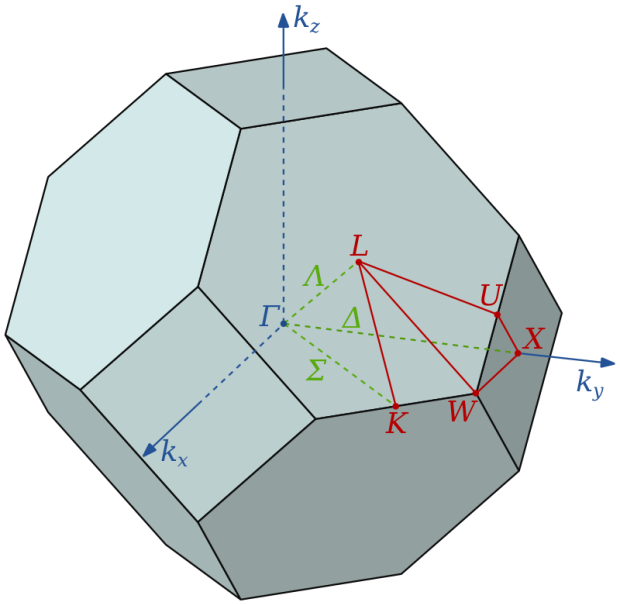
Acoustic and Optical Phonons



(Image: https://commons.wikimedia.org/wiki/File:Optical_%26_acoustic_vibrations.png)

Dispersion Relation in 3D

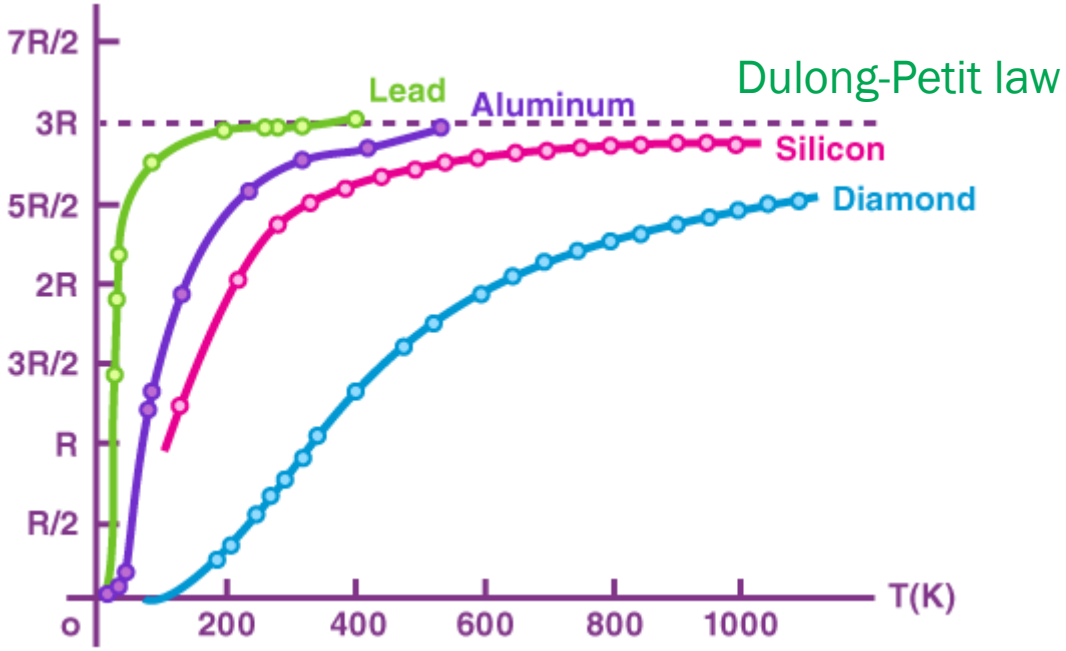




(Images: http://lampz.tugraz.at/~hadley/ss1/problems/problems_phonons.php
https://commons.wikimedia.org/wiki/File:Brillouin_Zone_(1st,_FCC).svg)

Heat Capacity of Solid

molar heat capacity



(Image: https://blog.naver.com/jinest/70153707595)

Classical Treatment

- $3N$ independent (identical) harmonic oscillators

$$Z = z_{\mathrm{vib}}^{3N}$$

$$z_{\mathrm{vib}} = \frac{1}{h}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} e^{-\beta\left(\frac{1}{2}kx^2+\frac{p^2}{2m}\right)}\,\mathrm{d}x\mathrm{d}p = \frac{k_{\mathrm{B}}T}{h\nu}$$

$$Z = \left(\frac{k_B T}{h\nu}\right)^{3N} = (\beta h\nu)^{-3N}$$

Heat Capacity

$$Z = (\beta h\nu)^{-3N}$$

$$U = -\frac{\partial \ln Z}{\partial \beta} = \frac{3N}{\beta} = 3Nk_{\mathrm{B}}T = 3nRT$$

Hence, the heat capacity is

$$C_V = \left(\frac{\partial U}{\partial T}\right)_V = 3nR$$

(Dulong-Petit law)

Improvement: Normal Modes

- $3N$ independent (different) harmonic oscillators

$$z_{\mathrm{vib},i} = \frac{k_{\mathrm{B}}T}{h\nu_i}$$

$$Z = z_{\mathrm{vib},1} z_{\mathrm{vib},2} \cdots z_{\mathrm{vib},3N} = \prod_{i=1}^{3N} z_{\mathrm{vib},i}$$

$$Z = \left(\frac{k_B T}{h}\right)^{3N} \frac{1}{\prod_{i=1}^{3N} \nu_i} = (\beta h)^{-3N} \frac{1}{\prod_{i=1}^{3N} \nu_i}$$

Heat Capacity

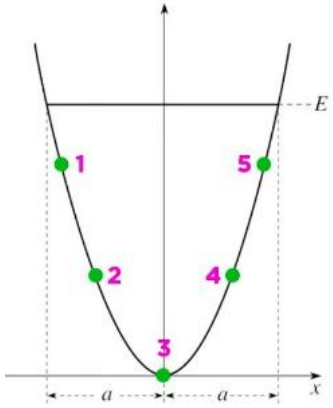
$$Z = (\beta h)^{-3N} \frac{1}{\prod_{i=1}^{3N} \nu_i}$$

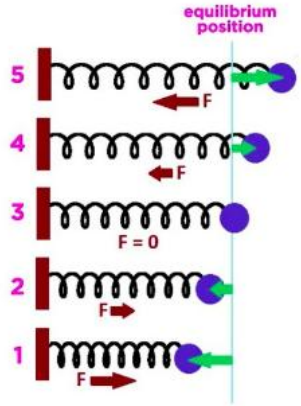
$$U = -\frac{\partial \ln Z}{\partial \beta} = \frac{3N}{\beta} = 3Nk_{\mathrm{B}}T = 3nRT$$

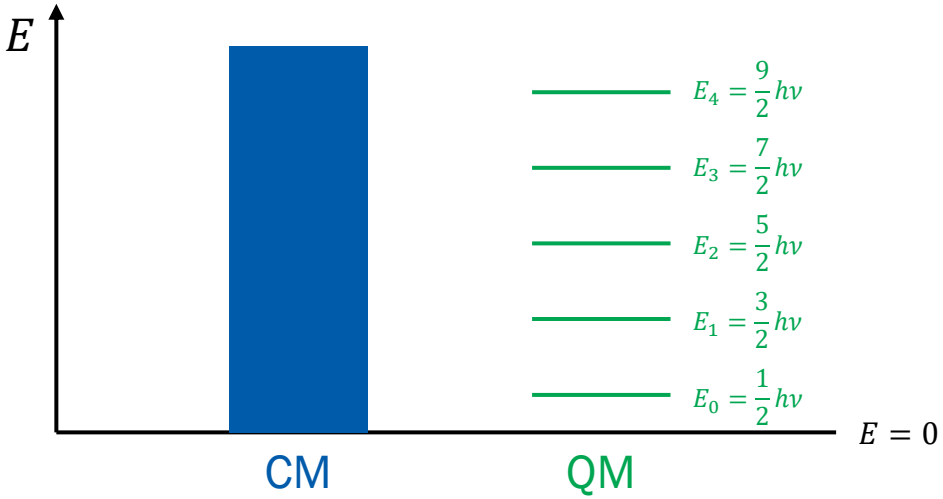
Hence, the heat capacity is (again)

$$C_V = \left(\frac{\partial U}{\partial T}\right)_V = 3nR$$

Quantization of Vibration





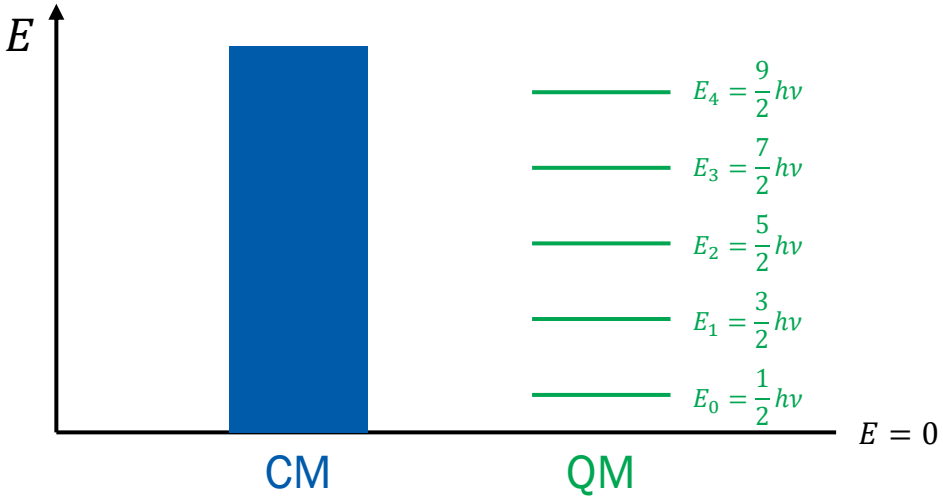


(Image: https://www.youtube.com/watch?v=yG_0t9rsNaw)

“Phonon”

A phonon is a discrete quantum of vibration

(bosonic)



Einstein Model

- $3N$ independent (identical) harmonic oscillators

$$Z = z_{\text{vib,QM}}^{3N}$$

Note: $\omega = 2\pi\nu$

$$z_{\text{vib,QM}} = \sum_{v=0}^{\infty} e^{-\beta\hbar\omega\left(v+\frac{1}{2}\right)} = \frac{e^{-\beta\hbar\omega/2}}{1-e^{-\beta\hbar\omega}}$$

$$Z = \left(\frac{e^{-\beta\hbar\omega/2}}{1-e^{-\beta\hbar\omega}}\right)^{3N}$$

Heat Capacity

$$Z = \left(\frac{e^{-\beta\hbar\omega/2}}{1 - e^{-\beta\hbar\omega}}\right)^{3N}$$

$$U = -\frac{\partial \ln Z}{\partial \beta} = \frac{\partial}{\partial \beta}\left[\frac{3}{2}N\hbar\omega\beta + 3N\ln\left(1 - e^{-\beta\hbar\omega}\right)\right]$$

$$U = \frac{3}{2}N\hbar\omega + 3N\hbar\omega\frac{e^{-\beta\hbar\omega}}{1 - e^{-\beta\hbar\omega}} = \frac{3}{2}N\hbar\omega + 3N\hbar\omega\frac{1}{e^{\beta\hbar\omega} - 1}$$

$$C_V = \left(\frac{\partial U}{\partial T}\right)_V = \frac{\mathrm{d}\beta}{\mathrm{d}T}\frac{\partial U}{\partial \beta} = \frac{1}{k_\mathrm{B}T^2}\cdot\frac{3N(\hbar\omega)^2 e^{\beta\hbar\omega}}{(e^{\beta\hbar\omega} - 1)^2}$$

Einstein's Prediction

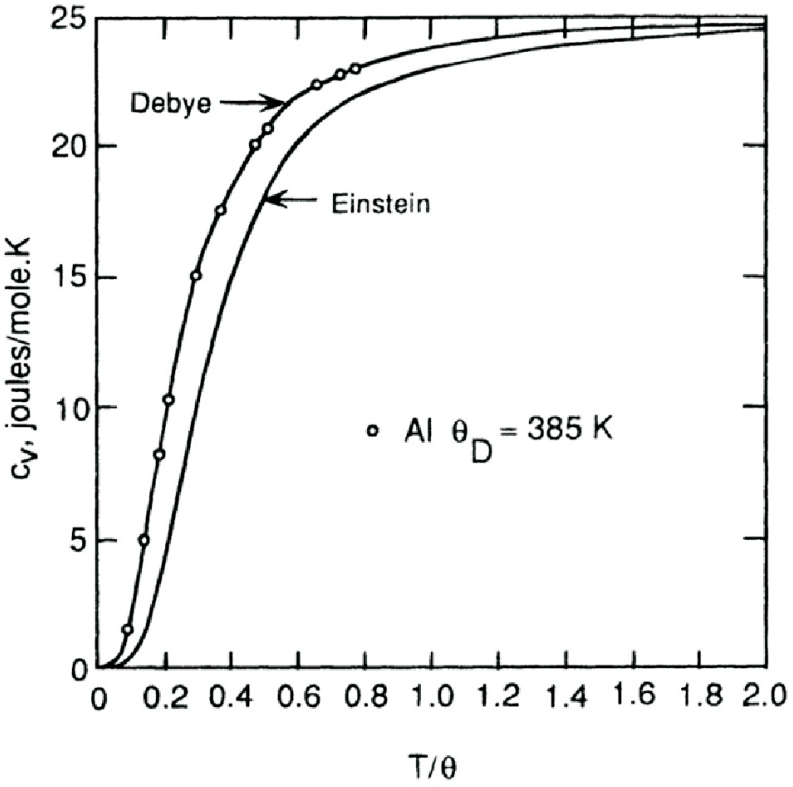
$$C_V = \frac{1}{k_B T^2} \cdot \frac{3N(\hbar\omega)^2 e^{\beta\hbar\omega}}{(e^{\beta\hbar\omega} - 1)^2}$$

$$C_V = 3Nk_B(\beta\hbar\omega)^2 \frac{e^{\beta\hbar\omega}}{(e^{\beta\hbar\omega} - 1)^2}$$

As $T \to 0$, $e^{\beta\hbar\omega} = e^{\hbar\omega/k_B T} \gg 1$ and

$$C_V \approx 3Nk_B(\beta\hbar\omega)^2 e^{-\beta\hbar\omega} \propto \frac{1}{T^2} e^{-\hbar\omega/k_B T}$$

Experimental Data



(Image: https://www.tf.uni-kiel.de/matwis/amat/td_kin_i/kap_1/backbone/r_su4.html)

Boson Counting

$$U = \frac{3}{2}N\hbar\omega + 3N\hbar\omega\frac{1}{e^{\beta\hbar\omega} - 1}$$

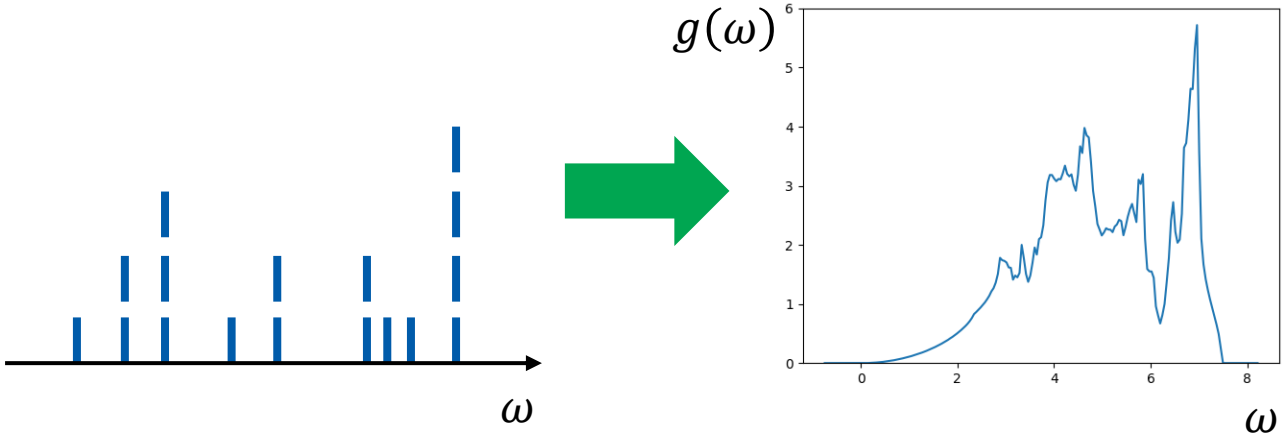
Ignoring the zero-point energy (constant),

$$U = 3N\hbar\omega\frac{1}{e^{\beta\hbar\omega} - 1} = 3N\hbar\omega N_{\mathrm{BE}}(\hbar\omega)$$

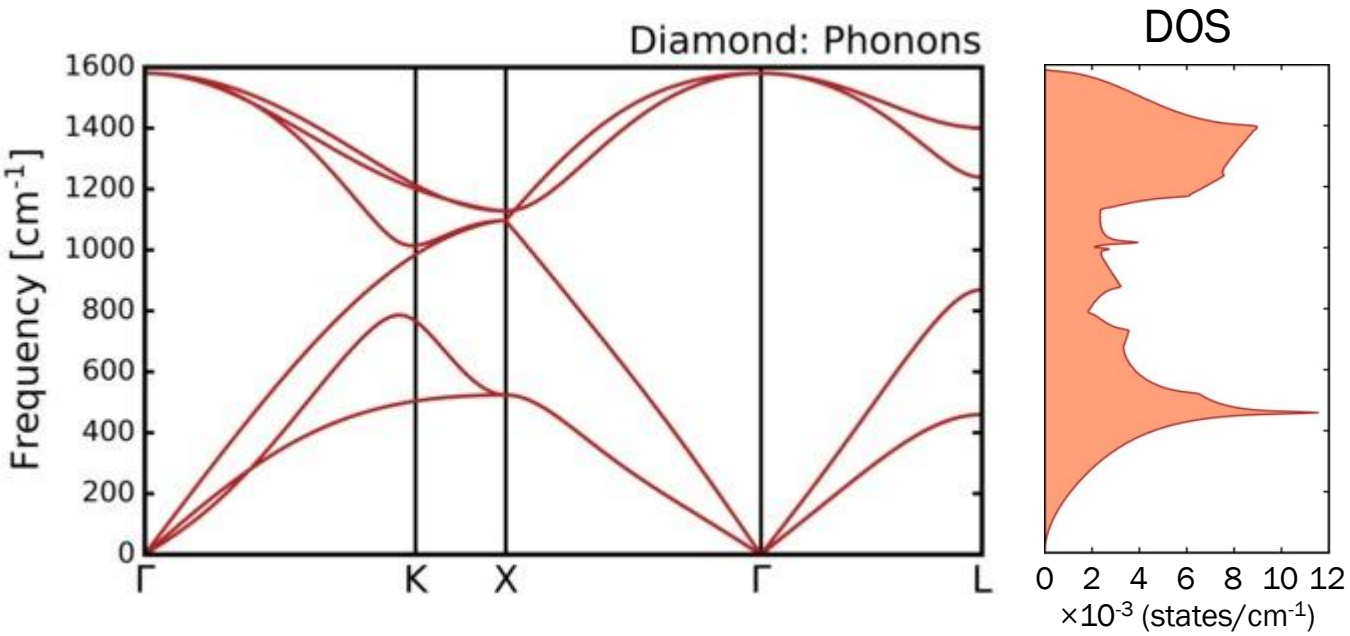
where $N_{\mathrm{BE}}(\hbar\omega)$ is the average occupation number of bosons for a microstate with energy $\hbar\omega$

Debye Model

- $3N$ independent (different) harmonic oscillators
- We have large N: many normal modes with different ω_i
- Consider them as continuous!
- Density of states (DOS) $g(\omega)$



Determination of DOS



(Image: https://www.tf.uni-kiel.de/matwis/amat/td_kin_i/kap_1/backbone/r_su4.html)

DOS of Simple Model

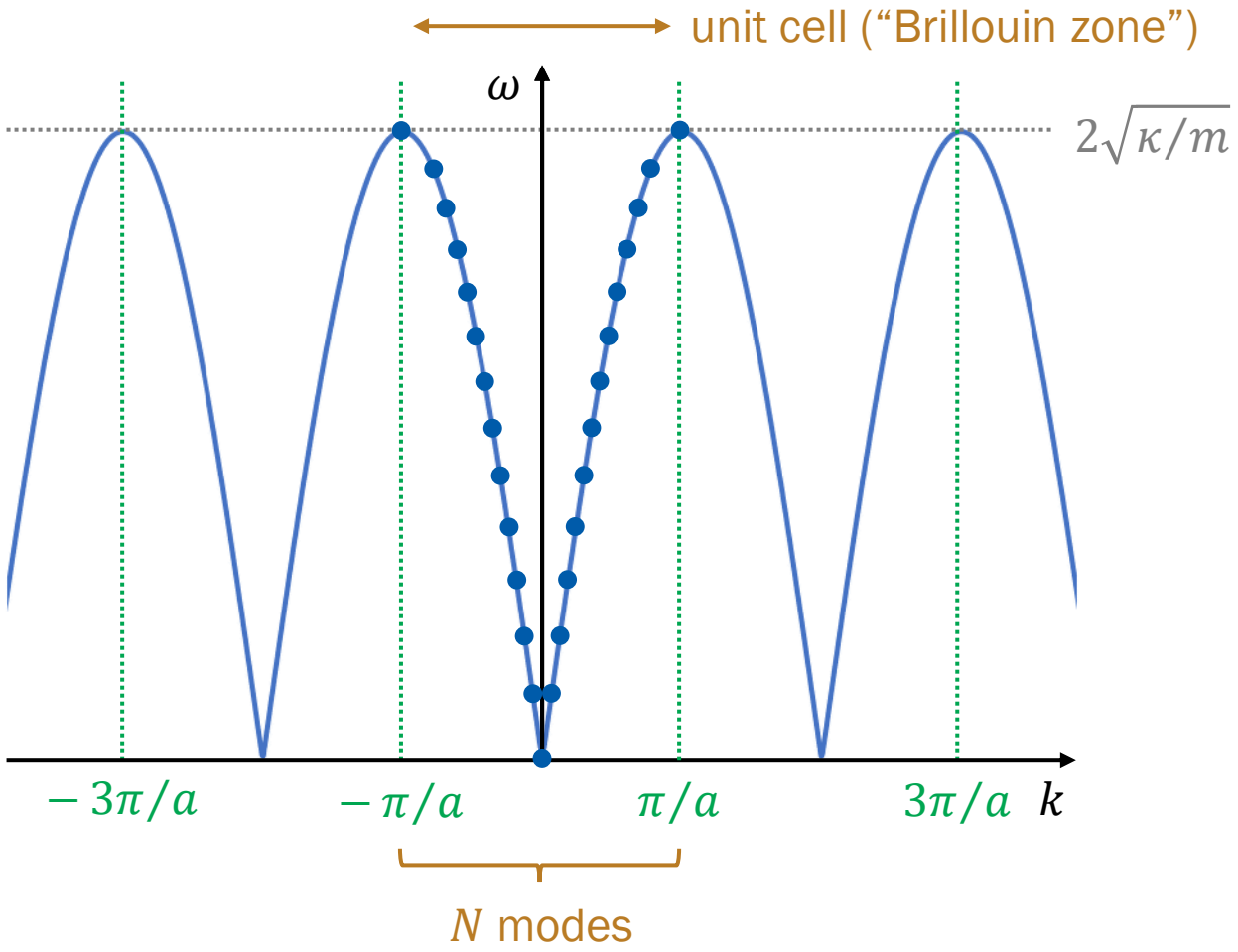
- $g(\omega)\,\mathrm{d}\omega = \mathrm{d}N_{\mathrm{modes}}$: number of modes in $[\omega, \omega + \mathrm{d}\omega]$
- $h(k)\,\mathrm{d}k = \mathrm{d}N_{\mathrm{modes}}$: number of modes in $[k, k + \mathrm{d}k]$

$$h(k) = \frac{\mathrm{d}N_{\mathrm{modes}}}{\mathrm{d}k}$$

How can we count the number of modes?

The dispersion relation is continuous only when $N \to \infty$; consider finite N (= finite system size)!

Case of 1D Simple Model



DOS Calculation

- $g(\omega)\,\mathrm{d}\omega = \mathrm{d}N_{\mathrm{modes}}$: number of modes in $[\omega, \omega + \mathrm{d}\omega]$
- $h(k)\,\mathrm{d}k = \mathrm{d}N_{\mathrm{modes}}$: number of modes in $[k, k + \mathrm{d}k]$

$$h(k) = \frac{\mathrm{d}N_{\mathrm{modes}}}{\mathrm{d}k}$$

$$h_{1\mathrm{D}}(k) \sim \frac{N}{2\pi/a} = \frac{L}{2\pi}$$

$$h_{3\mathrm{D}}(\vec{k}) \sim \frac{N^3}{(2\pi/a)^3} = \frac{L^3}{(2\pi)^3} = \frac{V}{(2\pi)^3}$$

DOS Calculation

- $g(\omega)\,\mathrm{d}\omega = \mathrm{d}N_{\mathrm{modes}}$: number of modes in $[\omega, \omega + \mathrm{d}\omega]$
- $h(k)\,\mathrm{d}k = \mathrm{d}N_{\mathrm{modes}}$: number of modes in $[k, k + \mathrm{d}k]$

$$h_{3\mathrm{D}}(\vec{k}) \sim \frac{N^3}{(2\pi/a)^3} = \frac{L^3}{(2\pi)^3} = \frac{V}{(2\pi)^3}$$

The number of modes in $[0, k]$ is then

$$N_{\mathrm{modes}}(k) = \int_{\mathrm{volume}} h_{3\mathrm{D}}(\vec{k})\,\mathrm{d}\vec{k} = \int_0^k h_{3\mathrm{D}}(\vec{k})\,4\pi k^2\,\mathrm{d}k$$

$$N_{\mathrm{modes}}(k) = \frac{Vk^3}{6\pi^2}$$

DOS Calculation

- $g(\omega)\,\mathrm{d}\omega = \mathrm{d}N_{\mathrm{modes}}$: number of modes in $[\omega, \omega + \mathrm{d}\omega]$

The number of modes in $[0, k]$ is

$$N_{\mathrm{modes}}(k) = \frac{Vk^3}{6\pi^2}$$

$$g(\omega) = \frac{\mathrm{d}N_{\mathrm{modes}}}{\mathrm{d}\omega} = \frac{Vk^2}{2\pi^2}\frac{1}{\mathrm{d}\omega/\mathrm{d}k}$$

Density of States

$$g(\omega) = \frac{Vk^2}{2\pi^2}\frac{1}{\mathrm{d}\omega/\mathrm{d}k}$$

Note: $\omega = 2\sqrt{\frac{\kappa}{m}}\left|\sin\left(\frac{ka}{2}\right)\right|$

Assuming a simple dispersion relation, $\omega = vk$,

$$g(\omega) = \frac{V\omega^2}{2\pi^2 v^3}$$

Considering longitudinal and transverse waves,

$$g(\omega) = g_{\mathrm{l}}(\omega) + 2g_{\mathrm{t}}(\omega) = \frac{V\omega^2}{2\pi^2 v_{\mathrm{l}}^3} + \frac{V\omega^2}{\pi^2 v_{\mathrm{t}}^3}$$

Energy Calculation

$$g(\omega) = g_{\mathrm{l}}(\omega) + 2g_{\mathrm{t}}(\omega) = \frac{V\omega^2}{2\pi^2 v_{\mathrm{l}}^3} + \frac{V\omega^2}{\pi^2 v_{\mathrm{t}}^3}$$

$$U = \int_0^{?} g(\omega)\hbar\omega N_{\mathrm{BE}}(\hbar\omega)\,\mathrm{d}\omega$$

Do we have the upper limit?

Upper Limit ω_c

No waves can propagate inside the solid

with a too small wavelength (= too high frequency)

$$\int_0^{\omega_c} g(\omega)\,\mathrm{d}\omega = 3N$$

$$\int_0^{\omega_c} \left(\frac{V}{2\pi^2 v_l^3} + \frac{V}{\pi^2 v_t^3}\right)\omega^2\,\mathrm{d}\omega = 3N$$

$$\omega_c^3 = 9N\left(\frac{V}{2\pi^2 v_l^3} + \frac{V}{\pi^2 v_t^3}\right)^{-1}$$

Energy Calculation

$$\omega_{\mathrm{c}}^3 = 9N\left(\frac{V}{2\pi^2 v_{\mathrm{l}}^3} + \frac{V}{\pi^2 v_{\mathrm{t}}^3}\right)^{-1}$$

$$g(\omega) = \frac{V\omega^2}{2\pi^2 v_{\mathrm{l}}^3} + \frac{V\omega^2}{\pi^2 v_{\mathrm{t}}^3} = \frac{9N\omega^2}{\omega_{\mathrm{c}}^3}$$

$$U = \int_0^{\omega_{\mathrm{c}}} g(\omega)\hbar\omega N_{\mathrm{BE}}(\hbar\omega)\,\mathrm{d}\omega$$

$$U = \frac{9N\hbar}{\omega_{\mathrm{c}}^3}\int_0^{\omega_{\mathrm{c}}} \omega^3 \frac{1}{e^{\beta\hbar\omega} - 1}\,\mathrm{d}\omega$$

(No closed form)

Heat Capacity

$$U = \frac{9N\hbar}{\omega_c^3}\int_0^{\omega_c} \omega^3 \frac{1}{e^{\beta\hbar\omega}-1}\,\mathrm{d}\omega$$

$$C_V = \left(\frac{\partial U}{\partial T}\right)_V = \frac{\mathrm{d}\beta}{\mathrm{d}T}\frac{\partial U}{\partial \beta} = \frac{1}{k_B T^2}\frac{9N\hbar}{\omega_c^3}\int_0^{\omega_c} \frac{\hbar\omega^4 e^{\beta\hbar\omega}}{(e^{\beta\hbar\omega}-1)^2}\,\mathrm{d}\omega$$

Letting $x = \beta\hbar\omega$ and $x_c = \beta\hbar\omega_c$,

$$C_V = \frac{9Nk_B}{x_c^3}\int_0^{x_c} \frac{x^4 e^x}{(e^x-1)^2}\,\mathrm{d}x$$

Debye T^3 Law

$$C_V = \frac{9Nk_\mathrm{B}}{x_\mathrm{c}^3}\int_0^{x_\mathrm{c}} \frac{x^4 e^x}{(e^x-1)^2}\,\mathrm{d}x$$

As $T \to 0$, $x_\mathrm{c} = \beta\hbar\omega_\mathrm{c} \to \infty$, and

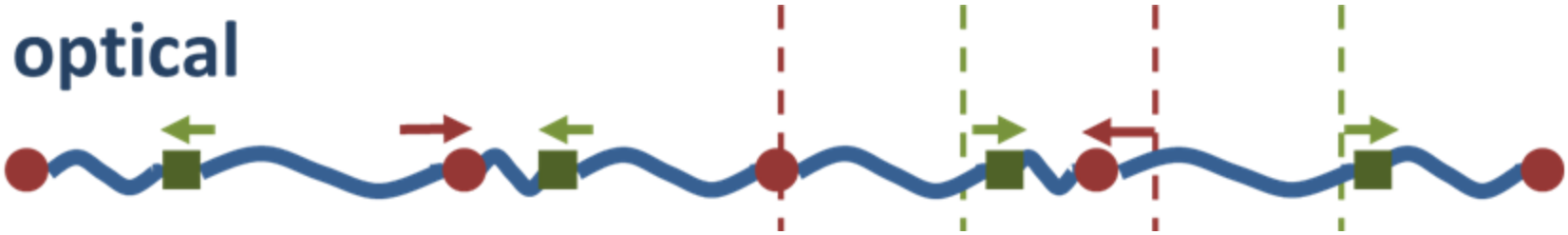
$$C_V \approx \frac{9Nk_\mathrm{B}}{x_\mathrm{c}^3}\int_0^{\infty} \frac{x^4 e^x}{(e^x-1)^2}\,\mathrm{d}x \propto \frac{1}{x_c^3} \propto \frac{1}{\beta^3} \propto T^3$$

Quantum: Selection Rules?

Vibrational Selection Rules

Dipole moment of the molecule must vary during a vibration

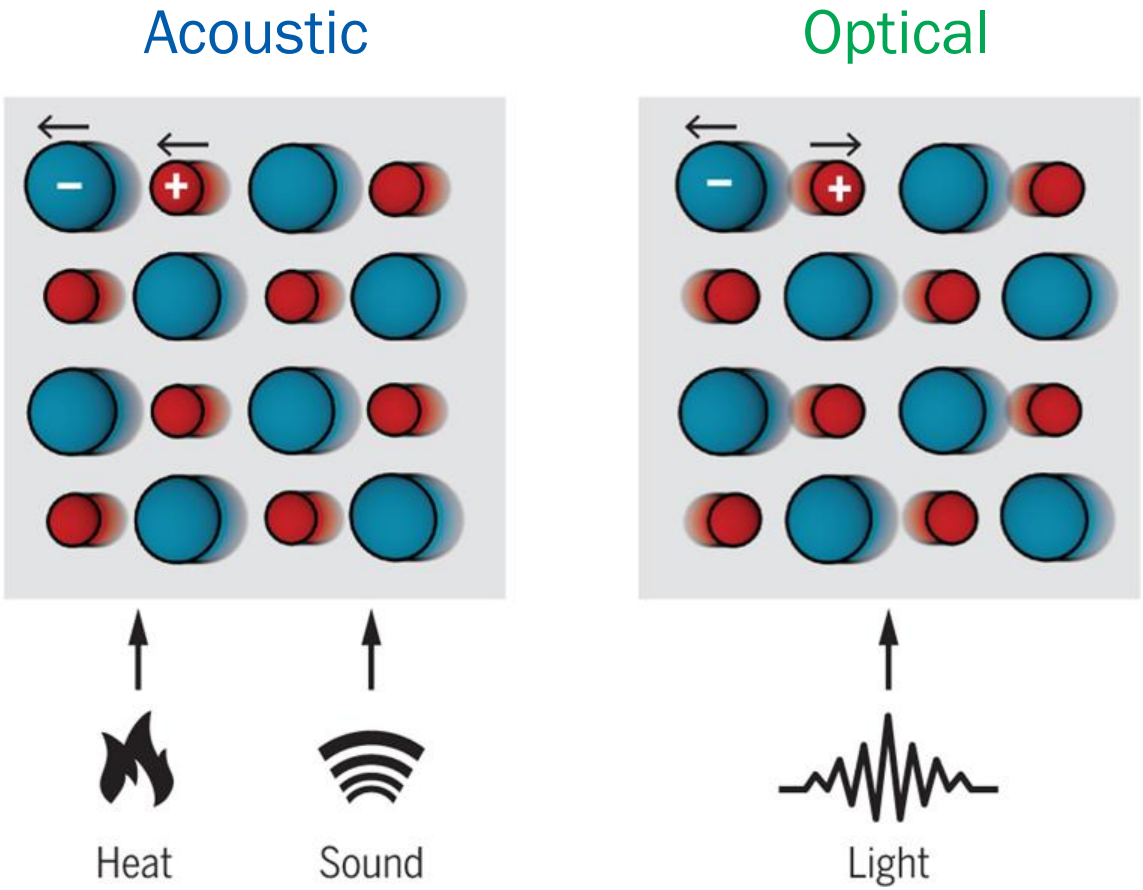
- Infrared active: the normal mode whose $\mathrm{d}\mu/\mathrm{d}q_j \neq 0$
- Infrared inactive: the normal mode whose $\mathrm{d}\mu/\mathrm{d}q_j = 0$



Optical phonons are IR active!

(Image: https://commons.wikimedia.org/wiki/File:Optical_%26_acoustic_vibrations.png)

Two Types of Phonons



(Image: Juraschek and Spaldin, ***Science*** (2017))